A Counterexample to the Clark Cauchy Transform Converse

Automated Math Discovery Smoke Test

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Source and Open Problem

This packet concerns Open Problem 4.4 from:

> C. Bellavita, E. Dellepiane, A. Hartmann, and J. Mashreghi, *Infinitely supported harmonically weighted Dirichlet spaces which are de Branges Rovnyak spaces*, arXiv:2509.04907.

The source proves that if u is a one-component inner function and σ is its Clark measure, then the truncated Cauchy transform

$$\mathcal{C}_\sigma f(\zeta) = \int_{\mathbb{T}\setminus\{\zeta\}} \frac{f(s)}{1 - \overline{s}\zeta}\, d\sigma(s)$$

is bounded on $L^2(\sigma)$. It then asks whether the converse is true:

> Open problem 4.4. Let σ be the Clark measure associated with u. Is it true that if $\mathcal{C}_\sigma$ is bounded on $L^2(\sigma)$, then u is a one-component inner function?

that $I_a \lesssim \sigma(2Q) \lesssim \sigma(Q)$. Hence, we get (4.1) for every arc Q. By applying Theorem 4.1, we conclude the proof. $\square$

We emphasize that the relationship between the inner function u and the Cauchy transform $\mathcal{C}_\sigma$ is far from being fully understood. In this regard, we conclude this section with an open problem.

Open problem 4.4. *Let σ be the Clark measure associated with u. Is it true that if $\mathcal{C}_\sigma$ is bounded on $L^2(\sigma)$, then u is a one-component inner function?*

If σ is not singular, we refer the reader to [8, Theorem 1.7] for the analogue of Theorem 3.8.

5. Proof of Theorem 2.3: Necessary conditions

In this section we start the proof of Theorem 2.3: we show that,

Idea of the Proof

Boundedness of the full Cauchy transform controls the singular integral operator, but it does not force the geometric mass-gap conditions appearing in Bessonov's characterization of Clark measures of one-component inner functions. We make the atom masses so small that the Cauchy matrix is Hilbert-Schmidt. At the same time, the masses are far too small compared to the adjacent gaps, so Bessonov's necessary lower estimate fails.

The Counterexample

Theorem 1. *There is an inner function u with Clark measure σ such that $\mathcal{C}_\sigma$ is bounded on $L^2(\sigma)$, but u is not one-component.*

Proof. For $n \geq 1$ put

$$\theta_n = 2^{-n}, \qquad \zeta_n = e^{i\theta_n}, \qquad m_n = 2^{-4n},$$

and define the finite positive singular measure

$$\sigma = \sum_{n=1}^{\infty} m_n \delta_{\zeta_n}.$$

By the standard Herglotz–Clark correspondence for singular measures on $\mathbb{T}$, there is an inner function u for which σ is the Clark measure corresponding to the parameter 1.

We first prove that $\mathcal{C}_\sigma$ is bounded on $L^2(\sigma)$. Identifying $L^2(\sigma)$ with $\ell^2(m_n)$, the transform is represented on finitely supported sequences by

$$(\mathcal{C}_\sigma f)(\zeta_j) = \sum_{i \neq j} \frac{m_i f(\zeta_i)}{1 - \overline{\zeta_i}\zeta_j}.$$

Conjugating by the unitary map $f \mapsto (m_n^{1/2} f(\zeta_n))_{n \geq 1}$ gives the matrix

$$a_{ji} = \begin{cases} \dfrac{(m_i m_j)^{1/2}}{1 - \overline{\zeta_i}\zeta_j}, & i \neq j, \\ 0, & i = j. \end{cases}$$

It is enough to show that this matrix is Hilbert-Schmidt.

For $i < j$,

$$|\theta_i - \theta_j| = 2^{-i} - 2^{-j} \geq 2^{-i-1}.$$

Since $|\theta_i - \theta_j| < 1$, the elementary estimate $|1 - e^{it}| \geq c|t|$ gives

$$|1 - \overline{\zeta_i}\zeta_j| = |1 - e^{i(\theta_j - \theta_i)}| \geq c\, 2^{-i}.$$

Therefore

$$|a_{ji}|^2 \leq C\, 2^{-4i} 2^{-4j} 2^{2i} = C\, 2^{-2i-4j} \qquad (i < j).$$

The same estimate with i and j interchanged handles $j < i$. Hence

$$\sum_{i,j \geq 1} |a_{ji}|^2 \leq 2C \sum_{1 \leq i < j} 2^{-2i-4j} < \infty.$$

Thus $\mathcal{C}_\sigma$ is Hilbert-Schmidt, in particular bounded, on $L^2(\sigma)$.

It remains to show that u is not one-component. The source quotes Bessonov's characterization as Theorem 3.8. One necessary condition for a Clark measure of a one-component inner function is the existence of a constant $A > 0$ such that every isolated atom ξ satisfies

$$\sigma(\{\xi\}) \geq A|\xi - \xi^+|$$

for each adjacent atom ξ^+. Here ζ_{n+1} is adjacent to ζ_n in the accumulating chain, and

$$|\zeta_n - \zeta_{n+1}| \asymp |\theta_n - \theta_{n+1}| = 2^{-n-1}.$$

Consequently

$$\frac{\sigma(\{\zeta_n\})}{|\zeta_n - \zeta_{n+1}|} \asymp \frac{2^{-4n}}{2^{-n}} = 2^{-3n} \longrightarrow 0.$$

No positive lower mass-gap constant A exists. Therefore Bessonov's necessary condition fails, and the inner function u cannot be one-component. □

Verification and Novelty Check

The file `code/check_hilbert_schmidt.py` computes finite partial sums of the Hilbert-Schmidt norm for the displayed matrix. This is only a sanity check; the proof above gives the exact summability estimate.

Before promotion, the run indexes were searched for `2509.04907`, the source title, `Clark measure`, `Cauchy transform`, and `one-component inner function`. Web searches for exact and nearby phrases from Open Problem 4.4 only found the source paper. The counterexample uses the problem exactly as stated; it does not address possible strengthened versions where one assumes Bessonov's atom-neighbor geometry or mass-gap conditions in advance.

References

References

[1] C. Bellavita, E. Dellepiane, A. Hartmann, and J. Mashreghi, *Infinitely supported harmonically weighted Dirichlet spaces which are de Branges Rovnyak spaces*, arXiv:2509.04907, 2025.

[2] R. V. Bessonov, *Duality theorems for coinvariant subspaces of H^1*, Advances in Mathematics **271** (2015), 62–90.

[3] J. A. Cima, A. L. Matheson, and W. T. Ross, *The Cauchy Transform*, Mathematical Surveys and Monographs, American Mathematical Society, 2006.